

Using Ant Colony Optimised Algorithms for Shortest Path Exploration by Robots

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Abstract— In the realm of robotics and autonomous systems, efficient path planning is a critical aspect for optimizing resource utilization and achieving mission objectives. This study explores the application of Ant Colony Optimized (ACO) algorithms to enhance the efficiency of robots in navigating and exploring unknown environments. Inspired by the foraging behavior of ants, ACO algorithms are employed to find the shortest paths by leveraging pheromone communication and decentralized decision-making. Ant Colony Optimization is then introduced as a bio-inspired solution to address these challenges. The experimental results demonstrate the capability of the ACO algorithm to efficiently explore complex environments, adapt to changes in the surroundings, and converge to optimal paths. Insights gained from these analyses contribute to the refinement of the ACO algorithm for diverse robotic applications. The findings of this research have practical implications for the development of autonomous robots operating in real-world scenarios, such as search and rescue missions, environmental monitoring, and surveillance. This study provides a valuable foundation for the integration of bio-inspired algorithms in robotics, advancing the state-of-the-art in autonomous exploration and path planning.

Keywords— Robot, Path Planning, Heuristic approach, ACO, Pheromone

I. INTRODUCTION

The field of robotics has witnessed significant advancements in recent years, with autonomous systems becoming integral to various applications such as search and rescue, surveillance, environmental monitoring, and logistics. A fundamental challenge faced by these robotic systems is efficient path planning in dynamic and unknown environments [1]. The ability to navigate through complex terrains while optimizing for the shortest path is crucial for achieving mission objectives, conserving energy, and ensuring the overall success of robotic operations.

Traditional path-planning algorithms often face limitations in adaptability to dynamic environments, scalability, and efficiency in finding optimal paths [2]. In response to these challenges, bio-inspired algorithms have gained prominence for their ability to mimic nature's solutions to complex problems. One such promising

approach is the application of Ant Colony Optimization (ACO) algorithms, inspired by the foraging behavior of ants.

Traditional algorithms, such as Dijkstra's algorithm and A* search algorithm, have been widely used for path planning in robotics. While effective in certain scenarios, these approaches may struggle with scalability and adaptability when faced with complex and dynamic environments. Their deterministic nature can lead to suboptimal solutions in situations where flexibility and adaptability are paramount.

Ant Colony Optimization, introduced by Marco Dorigo [3] in the early 1990s, is a nature-inspired metaheuristic that models the foraging behavior of ants to find optimal paths. Ants deposit pheromones on paths they traverse, and these pheromones influence the choices of other ants. This decentralized and stigmergic communication mechanism enables ants to collectively find the shortest paths between their nest and food sources [4]. The ACO algorithm adapts this concept to guide robots in exploring and navigating unknown environments. Previous research has demonstrated the effectiveness of ACO algorithms in various robotics applications, including mobile robot navigation, swarm robotics, and multi-agent systems. The decentralized nature of ACO allows robots to adapt to dynamic changes in the environment and find optimal paths without requiring a centralized control mechanism.

The review reveals a growing body of research showcasing the effectiveness of ACO in enhancing path-planning capabilities for robots. The decentralized, nature-inspired approach of ACO aligns well with the challenges posed by dynamic and unknown environments, positioning it as a valuable tool for autonomous robotic systems [5]. Ongoing research focuses on addressing challenges and expanding the applicability of ACO, further solidifying its role in the field of robotics. Ant Colony Optimization (ACO) for path planning involves several steps to find optimal or near-optimal paths through a given environment [6]. Here are the general steps for implementing ACO for path planning:

- Clearly define the path planning problem, including the characteristics of the environment, the start and goal locations, and any obstacles or constraints that need to be considered.

- Represent the environment as a graph, where nodes represent locations, and edges represent potential paths between locations. The graph should capture the connectivity and distances between different points in the environment.
- Establish an objective function that the algorithm aims to optimize. In path planning, the objective function is typically the total cost or distance of the path. This function guides the ants in their exploration to find shorter paths.
- Assign initial pheromone levels to all edges in the graph. The pheromone level represents the desirability of a particular path. Higher pheromone levels indicate a more attractive path. Initialize a population of artificial ants. Each ant is placed at the starting point of the environment and is tasked with finding a path to the goal.
- Allow each ant to construct a path from the starting point to the goal by iteratively moving from one node to another. The ant's decision-making process is influenced by both pheromone levels and a heuristic function (e.g., distance to the goal). Evaluate the paths constructed by each ant using the defined objective function.
- Update the pheromone levels on the edges based on the quality of the paths. Shorter paths receive higher pheromone deposits. Simulate the natural evaporation of pheromones over time. This prevents the algorithm from converging too quickly to suboptimal solutions and encourages the exploration of alternative paths.
- Define a termination condition, such as a maximum number of iterations or reaching a satisfactory solution. The algorithm stops when this condition is met. Extract the best path found by the ants at the end of the algorithm as the solution to the path planning problem.

These steps provide a general framework for implementing ACO for path planning. Depending on the specific requirements and characteristics of the problem, additional considerations and adaptations may be necessary. The effectiveness of ACO often depends on careful parameter tuning and a thorough understanding of the application domain.

Section II represents the literature survey to describe the findings in similar research areas. Section III describes the steps for the ACO algorithm for the planning of the shortest path between two end nodes. This section also gives the flowchart for the ACO algorithm. Section IV discusses the results based on average computation time and standard deviation in time. Section V finishes with the conclusion preceding references.

II. LITERATURE REVIEW

Yogita Gigras, Kusum Gupta [7] gave a backtracking method for avoiding collision and found the optimum shortest path to reach the destination by use of the ant colony algorithm. Buniyamin N. et al. [8] proposed a method to solve the Robot Path Planning (RPP) problem. They introduced the heuristic and visibility equations for transition rules of states. The new method was applied within a global

static map having feasible free space nodes. Michael Brand et al. [9] proposed an application of Ants Colony to planning the path of a robot in a dynamic environment. Two different pheromone re-initialization techniques were compared and designated the best shortest path based on the experimental result. O. Hachour [10] introduced a path-planning method in an unknown environment for an autonomous mobile robot. The robots move inside the environment by identifying and evading hindrances that appear crossways their way while traveling towards the destination station. Daniel Angus [11] improved the present ACO meta-heuristic technique by introducing three different parameters: cost, visibility, and pheromone. The new protocol for the Shortest Path Ant Colony Optimization (SPACO) was designed based on the three parameters. Visibility is a new and important factor that was added to this algorithm to solve the problem of the shortest path. M. Dorigo and C. Blum [12] proposed A Survey to discuss the theoretical results of Ant Colony Optimization algorithms. They examined the convergence results and the linking between random gradient ascent and the ACO algorithm within the model-based search. Kaur A. [13] et al. applied the ACO algorithm in to the multipath environment of AOMDV to find the optimized routes. If one path fails then the agents choose another available path to continue uninterrupted communication among source and destination nodes.

Ant Colony Optimisation is an optimized intelligent algorithm that has power to use the heuristic mechanism to identify the shortest route by using the pheromone while searching the food by ants [14]. ACO is also used in planning the paths of mobile robot. Akbarimajd et al. [15] proposed a path planning technique for mobile robots by using two-dimensional cellular automata that can be put on environments with both concave and convex obstacles. It can also be applied to multirobot problems in the dynamic environments. Montiel-Ross et al. [16] designed the software based on navigation which is called as Ant Colony Test Center. This software works on ant colonies and has the ability to achieve path generation, path planning, and virtual path tracking.

III. ANT COLONY OPTIMISATION ALGORITHM FOR SHORTEST PATH PLANNING

Create the graph of the existing path by considering each path as edge of graph and the intersection of path as nodes. Let L_k is the length of the k's path, ρ is evaporation constant whose value lies between $\rho \in (0,1)$, $\Delta\tau^k$ is increment in pheromone quantity whose value is $\left(\frac{1}{L_k}\right)$, N_i^k is neighbourhood of ant k when it is on the node i [17] [18]. α is a constant whose value is taken as 2 for experimental purposes. The following steps are needed to be executed.

1. Create N ants. Place them all on source node. Deposit a constant amount of pheromone $\tau_{ij} = 1$ on all the edges.
2. For Number of Iterations = 1 to S do,
 - for each ant do
 - Select an out-going edge from the current node where ant is with probability

$$p_{ij}^k = \begin{cases} \frac{\tau_{ij}^\alpha}{\sum_{i \in N_i^k} \tau_{ij}^\alpha} \\ 0, \text{ if } j \notin N_i^k \end{cases} \quad (i)$$

If more than one edge has the same amount of pheromone choose one randomly [15].

- Record this ant took this edge.
- If the end of the edge is the terminal node then
- Update the pheromone trail of the path traversed by ant k on each edge traversed by this ant. Record the path taken by the ant in the path table

$$\tau_{ij} \leftarrow \tau_{ij} + \left(\frac{1}{L_k}\right) \quad (ii)$$

- for each edge in the graph, evaporate the pheromone trail using the equation

$$\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij} \quad (iii)$$

- Reset the ant back to the initial state, clear its edge history and let it move as before. (end of “for each ant” loop)

3. At End of “Iteration step 1 to S do”

(loop counter ← loop counter + 1),
if not end of loop then go to step 2).

4. If the number of time steps are reasonable then all the ants start moving on the shortest path.

The procedure of traditional simple ACO [19] [20] is a kind of iterative process, and its flowchart is as follows shown in Fig 1.

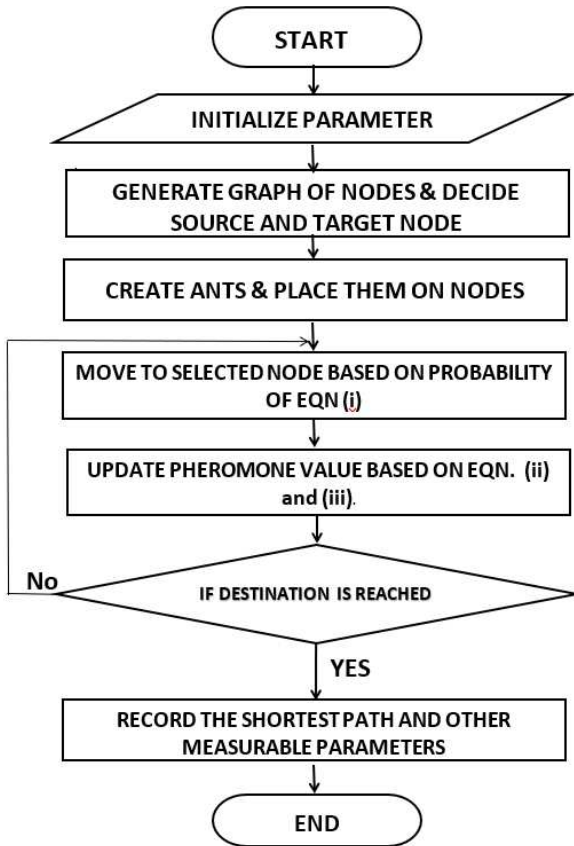


Fig. 1. Flowchart of the ACO algorithm

IV. RESULTS AND DISCUSSIONS

Let us consider example 1 for evaluating the results and further discussions.

Example 1: Let us consider a weighted directional graph for the movements of the robots as shown in Fig. 1.

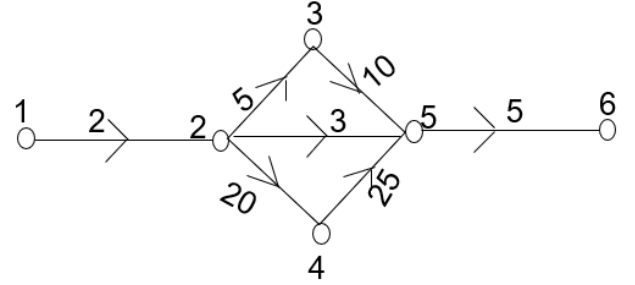


Fig. 2. 6-Node graph for the calculation of shortest path by Robots using S-ACO

	1	2	3	4	5	6
1	0	2	0	0	0	0
2	0	0	5	20	3	0
3	0	0	0	0	10	0
4	0	0	0	0	25	0
5	0	0	0	0	0	5
6	0	0	0	0	0	0

Fig. 3. Two-dimension node adjacency matrix

Now apply the simple ACO algorithm on the above 6-node graph. For this purpose, we need to construct a 6 x 6 two-dimensional matrix (because there are 6 nodes in the graph) as shown in Fig. 3 for the calculations. The experimental study of S-ACO on the above network results in the shortest path= 1-2-5-6.

On executing the A*, Dijkstra, and proposed algorithm 5 times, the average values are noted in terms of computation time and standard deviation which are represented in Table 1.

Table 1: Comparison Table for Average Results.

Name of Algorithm	Average Computation Time (milliseconds)	Standard Deviation of Time (milliseconds)
A* Algorithm	0.83652	0.00674
Dijkstra Algorithm	0.85236	0.00692
ACO based Algorithm	0.79964	0.00438

V. CONCLUSION

In conclusion, the Ant Colony Optimization algorithm has proven to be a valuable tool for robot shortest-path planning, offering a balance between exploration and

exploitation, adaptability to dynamic environments, and scalability. As research in this field continues, addressing challenges and refining the algorithm will likely enhance its performance and broaden its applicability in diverse robotic applications. The proposed methodology transforms the environment into a graph representation, with nodes representing locations and edges denoting potential paths. ACO through its inspiration from the foraging behavior of ants, ACO has demonstrated the ability to find optimal or near-optimal paths in complex and dynamic environments. The findings presented in this paper contribute to the growing body of research in autonomous robotics, providing a robust and nature-inspired solution for real-time path-planning challenges. As the demand for efficient autonomous navigation systems continues to grow, the integration of ACO in robot path planning stands out as a promising and effective strategy. The proposed algorithm proved better when compared with A* and Dijkstra Algorithm in terms of Average Computation Time and Standard Deviation of Time.

REFERENCES

- [1] Ridhi Bansal, Mojtaba Ahmadi Khanesar, David Branson, "Ant Colony Optimization Algorithm for Industrial Robot Programming in a Digital Twin", 25th International Conference on Automation and Computing (ICAC)-IEEEExplore, INSPEC Accession Number: 19136700, November 2019.
- [2] X. Jiang, Z. Lin, T. He, X. Ma, S. Ma, and S. Li, "Optimal path finding with beetle antennae search algorithm by using ant colony optimization initialization and different searching strategies," *IEEE Access*, vol. 8, no. 12, pp. 15459–15471, 2020.
- [3] M. Dorigo, G. Di Caro, Ant colony optimization: a new meta-heuristic. In: *Proceedings of the 1999 Congress on Evolutionary Computation (CEC 99)*, Vol. 2, 1999.
- [4] Gurpreet Singh, Baninder Singh, Dinesh Kumar, Amanpreet Kaur, Anil Kumar, "Applications of Ant Colony Optimization in the sustainable energy environment development", *Proceedings of National Conference on Recent Advances in the Development of Sustainable Environment-RADSE 2008*, pp 163-169, 3-4 May 2008.
- [5] Sharma S., Singh M. Singh G., "Simulated proposed finer algorithm based on Ant colony optimization in wired networks", *Yamuna Journal of Technology and Business Research*, Volume 2, Issue 1, pages 57-60, June 2012.
- [6] Singh A., Dhaka V. K., Singh G., "Ant Colony Optimization – A Prologue", *The International Journal of Advanced Networking and Applications (IJANA)*, Special Issue, Pages 7-11, 2015.
- [7] Yogita Gigras, Kusum Gupta, "Artificial Intelligence in Robot Path Planning", *International Journal of Soft Computing and Engineering (IJSCE)* ISSN: 2231-2307, Volume-2, Issue-2, May 2012.
- [8] Buniyamin N., Sariff N., Wan Ngah W.A.J., Mohamad Z., "Robot global path planning overview and a variation of ant colony system algorithm", *International Journal of Mathematics and Computers In Simulation*, vol 5, Issue 1, Pages 9-16, 2011.
- [9] Michael Brand, Michael Masuda, Nicole Wehner, Xiao-Hua Yu, "Ant Colony Optimization Algorithm for Robot Path Planning", 2010 International Conference on Computer Design And Applications (ICCD 2010), Vol. 3, INSPEC Accession Number: 11523768, pages 1-5, 2010.
- [10] O.Hachour "Path Planning of Autonomous Mobile Robot", *International journal of systems applications, engineering & development*, Issue 4, Volume 2, 2008
- [11] Daniel Angus, "Solving a unique Shortest Path problem using Ant Colony Optimization", Communicated by T.Baeck, https://www.academia.edu/972646/Solving_a_unique_Shortest_Path_problem_using_Ant_Colony_Optimisation.
- [12] Marco.Dorigo & Christian Blum , "Ant Colony Optimization Theory : A Survey", *Theoretical Computer Science*, Vol. 344, Issues 2–3, Pages 243-278, 17 November 2005.
- [13] Kaur A., Dhaka V. S., Singh G., "ACO Agent Based Routing in AOMDV Environment", *International Conference on Advancements in Engineering & Technology-2016 (ICAET-2016)*, pages 1-8, 18th & 19th March, 2016.
- [14] M. Dorigo, M. Birattari, and T. Stutzle, "Ant colony optimization," *IEEE Computational Intelligence Magazine*, Vol. 1, No. 4, pp. 28–39, 2006.
- [15] A. Akbarimajd and A. Hassanzadeh, "Autonomously implemented versatile path planning for mobile robots based on cellular automata and ant colony," *International Journal of Computational Intelligence Systems*, vol. 5, no. 1, pp. 39–52, 2012.
- [16] O. Montiel-Ross, R. Sepúlveda, O. Castillo, and P. Melin, "Ant colony test center for planning autonomous mobile navigation," *Computer Applications in Engineering Education*, vol. 21, No. 2, pp. 214–229, 2013.
- [17] Sharma S., Singh M. Singh G. "Realistic inspection of proposed Ant algorithm with Antnet algorithm using NS-2", *International Journal of Research in IT, Management and Engineering*, Volume 2, Issue 6, pp 146-156, June 2012.
- [18] Abu Saleh Bin Shahadat, M. A. H. Akhand, and Md Abdus Samad Kamal, "Visibility Adaptation in Ant Colony Optimization for Solving Traveling Salesman Problem", *Mathematics*, MDPI, Vol. 10, 2448, 2022.
- [19] Hongliu Huang, Guo Tan, Linli Jiang, "Robot Path Planning Using Improved Ant Colony Algorithm in the Environment of Internet of Things", *Journal of Robotics - Hindawi*, Col 2022, pp 1-8, April 2022.
- [20] Kaur A., Singh G., Singh A., Gupta R., Singh G., "Ant-Based Algorithm for Routing in Mobile Ad Hoc Networks", *Lecture Notes in Networks and Systems*, Springer, Vol 731, pp 357-366, 2024.